15th May 2026

#25093288
Yves Muhoza

<u>Assignment #3</u>

Computer graphics is the branch of computer science that deals with
- Creating
- Storing
- Manipulating
- Displaying images
- Designs
- Animations
- Visual content

Using a computer.
It combines art, mathematics, and technology to produce visual representations for communication, entertainment, education, engineering, medicine, and many other fields.

Computer graphics help users visualize information clearly and interact with digital content effectively.

Types of Computer Graphics

1. **Bitmap Graphics**
   Are made up of tiny dots called pixels, Each with contains color information
   Characteristics
   - Quality decreases when enlarged
   - Best for realistic images and photographs

   E.g:
   - Web graphics, digital photographs
2. **Vector Graphics**
   Are created using mathematical equations such as lines, curves, and shapes
   Characteristics
   - Smaller file size for simple designs
   - Can be resized without losing quality

   E.g:
   - Logos, Icons, Technical drawings
3. **2D Graphics**
   2D graphics represent objects in two dimensions: width and height
   Characteristics:
   - Flat images

- Used in diagrams, charts, and illustrations

E.g:

- Maps, UI

4. **3D Graphics**
   3D graphics represent objects with width, height, and depth
   Characteristics:
   - Realistic appearance
   - Uses lighting, shading, and textures
5. **Interactive Graphics**
   Allow users to control or manipulate graphical objects
   Applications:
   - Video games
   - Simulations
   - CAD systems
6. **Non-Interactive Graphics**
   These graphics are static and do not require user interaction
   Application:
   - Printed posters
   - Digital photos
   - Presentation slides

30 Reasons Why Computer Graphics Are Used and Their Applications

	Reason for using computer graphics	**Applications**
1.	Visual communication	Posters, presentation, advertisement
2.	Entertainment	Movies, animation, games
3.	Education and training	E-learning, virtual classrooms

4.	Scientific visualization	Weather forecasting, research simulations
5.	Medical imaging	X-rays, MRI scans
6.	Engineering design	CAD, Archad
7.	Architectural design	Building plans and 3D structures
8.	User interface design	Mobile apps, websites, software
9.	Advertising and marketing	Digital banners, commercials
10.	Data visualization	Charts, graphs, dashboards
11.	Simulation	Flight simulators
12.	Virtual reality	VR games and training systems
13.	Augmented reality	AR learning and shopping apps

14.	Image processing	Photo editing and enhancement
15.	Digital publishing	E-books, magazines
16.	Animation production	Cartoons and animated films
17.	Web development	Website graphics and interfaces
18.	Geographic information systems	Digital maps and navigation
19.	Gaming industry	2D and 3D game development
20.	Product design	Automobile and electronics modeling
21.	Fashion design	Virtual clothing design
22.	Artificial intelligence visualization	Ai-generated images and models
23.	Business presentations	Reports and visual summaries

24.	Social media communication	Memes, posts, digital campaigns
25.	Industrial automation	Machine control displays
26.	Film special effects	CGI effects in movies
27.	Robotics	Robot vision systems
28.	Digital art creation	Graphic design and illustration
29.	Security and surveillance	Facial recognition systems
30	Space and defence research	Satellite imaging and military simulations